

“Electric Vehicles and the Energy Transition: Unintended Consequences of Time-of-Use Pricing”

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Overview

The code in this replication package can be used to clean and process the raw data files, as well as to carry out the analyses. The data were provided by FortisAlberta and Optiwatt, our utility and industry partners, respectively. All programs are provided in the “Programs” folder. All of the Tables and Figures from the paper are generated within the code and are stored in the “Results” folder. Any source data files that are used are provided as template files in the “Data” folder with a data codebook summarized below. The data are stored in excel, csv, or dta STATA files. The programs used in the analysis are R and STATA. The code takes approximately 1-2 hours to run and generate the results from the paper.

Data Availability and Provenance Statements

Data Availability Statement:

The data in our paper are subject to a non-disclosure agreement with our utility and industry partners, FortisAlberta and Optiwatt. We are secondary users of the data collected from a program run by FortisAlberta. Supplementary data on US-based Optiwatt users were also provided directly by Optiwatt.

We provide data template files that reflect the data structure used in our analysis, including all of the variable labels and a brief description of what the data contain (see “Read Me Data template.xlsx”). However, the data has been removed due to confidentiality. Access to the data from our field experiment can be requested by contacting our utility partner FortisAlberta. Supplemental data on US EV users can be requested by contacting Optiwatt. Contacts: FortisAlberta Privacy Officer (privacyofficer@fortisalberta.com) and Todd Touesnard (todd@optiwatt.com).

The data will be preserved for 5 years in a restricted-access location. The authors commit to providing reasonable assistance to requests for clarification and replication over this time period.

Statement about Rights

- I certify that the authors of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Summary of Availability

- **No data can be made** publicly available. The data will be preserved for 5 years in a restricted-access location. Details on data availability and access are provided above.

Dataset list

In the “Data” folder, we have provided template data files that were used in our analysis. In the Excel file “Read Me Data template.xlsx”, we provide a summary of all variables in each data set. There is a sheet for each data set used in our analysis. The data source files used in our analysis are as follows:

1. Representative residential load – metered volumes data. Used to construct transformer capacity limits.
 - a. Data/Meter Data/AMI raw data extracts - July 1 2022 to March 1 2023.csv
2. EV Charging Files (Vehicle and Charge Event Sessions data) – these data include data on vehicle characteristics and a row for each charge session.
 - a. Data/Fortis Data/Fortis Data Transfer - 06_19_2023 UCalgary.xlsx
 - i. *Sheet:* Vehicle Data, Charge Event Data
 - b. Data/Fortis Data/Fortis Data Transfer - 07_07_2023 UCalgary.xlsx
 - i. *Sheet:* Vehicle Data
 - c. Data/Fortis Data/Fortis Data Transfer - 07_28_2023 UCalgary.xlsx
 - i. *Sheet:* Vehicle Data
 - d. Data/Fortis Data/Fortis Data Transfer - 08_22_2023.xlsx
 - i. *Sheet:* Vehicle Data
 - e. Data/Fortis Data/Fortis Data Transfer - 09_06_2023 - UCalgary.xlsx
 - i. *Sheet:* Vehicle Data
 - f. Data/Fortis Data/Fortis Data Transfer - 09_13_2023 - UCalgary.xlsx
 - i. *Sheet:* Vehicle Data
 - g. Data/Fortis Data/Fortis Data Transfer - 10_10_2023 - UCalgary.xlsx
 - i. *Sheet:* Vehicle Data
 - h. Data/Fortis Data/Fortis Data Transfer - 2023_11_08 UCalgary.xlsx
 - i. *Sheet:* Vehicle Data
 - i. Data/Fortis Data /Fortis Data Transfer - 2023_11_29 – UofC.xlsx
 - i. *Sheet:* Vehicle Owner Data
 - j. Data/Fortis Data/Fortis Data Transfer - 2023_12_18 UCalgary.xlsx
 - i. *Sheet:* Vehicle Data
 - k. Data/Fortis Data/Fortis Data Transfer - 2024_01_29 UCalgary.xlsx
 - i. *Sheet:* Vehicle Owner Data, Vehicle Data, Charge Event Data, Event Opt-outs

- I. Data/Fortis Data/Fortis - Data Transfer 2024-05-21 UCalgary.xlsx
 - i. *Sheet*: Vehicle Owner Data
3. EV Charging Interval data – detailed time-stamped data on EV charging.
 - a. Data/Fortis Data/Fortis Interval Data - 06_19_2023 UCalgary.csv
 - b. Data/Fortis Data/Fortis - Interval Charger Data 2024-01-29.csv
4. List of non-controllable or unverified EVs (that couldn't join program). Used in randomization to isolate sample for randomized assignment.
 - a. Data/Non-controllable vehicles-6-21-2023.xlsx
 - b. Data/Unverified EVs List-6-26-2023.xlsx
5. Locations of EVs in experiment and definition of urban/rural from Statistics Canada. Used to define % urban/suburban versus rural in our sample.
 - a. Data/Vehicle_List_LocationID -6-24-2024 - GC Value Update.xlsx
 - b. Data/community_data.csv
6. Panel data set of US EV charging patterns – Provided by Optiwatt.
 - a. Data/Optiwatt US/charge_sessions_hourly_panel.dta
7. Phase 3 survey – offering enrollment in subsequent managed charging program in December 2023. Includes list of EVs invited to the Phase 3 survey and the incentive offered and the file matching survey response to home id.
 - a. Data/Phase3_Incentive_vehicle_ids.dta
 - b. Data/Phase3_EligibleEVList.dta
 - c. Data/Fortis Data/Phase 3 Survey Data/\$0_V1_Match.dta
 - d. Data/Fortis Data/Phase 3 Survey Data/\$75_V1_Match.dta
 - e. Data/Fortis Data/Phase 3 Survey Data/\$150_V1_Match.dta

Details on each Data Source

The Table summarizes the data sets used in our analysis, their location in the replication package, and who provided us with the data source.

Name	Data.Files	Location	Provided	Provided By
Representative Load Metered Volumes	AMI raw data extracts - July 1 2022 to March 1 2023.csv	Data/Meter Data/	FALSE	FortisAlberta (2023)
Vehicle Info, Charge Sessions data	Fortis Data Transfer - 06_19_2023 UCalgary.xlsx Fortis Data Transfer - 07_07_2023 UCalgary.xlsx Fortis Data Transfer - 07_28_2023 UCalgary.xlsx Fortis Data Transfer - 08_22_2023.xlsx Fortis Data Transfer - 09_06_2023 - UCalgary.xlsx Fortis Data Transfer - 09_13_2023 - UCalgary.xlsx Fortis Data Transfer - 10_10_2023 - UCalgary.xlsx Fortis Data Transfer - 2023_11_08 UCalgary.xlsx Fortis Data Transfer - 2023_11_29 – UofC.xlsx Fortis Data Transfer - 2023_12_18 UCalgary.xlsx Fortis Data Transfer - 2024_01_29 UCalgary.xlsx Fortis - Data Transfer 2024-05-21 UCalgary.xlsx	Data/Fortis Data	FALSE	FortisAlberta (2023)
Charge Interval data	Fortis Interval Data - 06_19_2023 UCalgary.csv Fortis - Interval Charger Data 2024-01-29.csv	Data/Fortis Data	FALSE	FortisAlberta (2023)
Unverified and	Non-controllable vehicles-6-21-	Data	FALSE	FortisAlberta

Non-Controllable EVs	2023.xlsx Unverified EVs List-6-26-2023.xlsx			(2023)
EV Locations	Vehicle_List_LocationID -6-24-2024 - GC Value Update.xlsx community_data.csv	Data	FALSE	FortisAlberta (2023)
US EVs Panel	charge_sessions_hourly_panel.dta	Data/Optiwatt US	FALSE	Optiwatt (2024)
Phase 3 Survey – EV Lists	Phase3_Incentive_vehicle_ids.dta Phase3_EligibleEVList.dta	Data	FALSE	FortisAlberta (2023)
Phase 3 Survey Results	\$0_V1_Match.dta \$75_V1_Match.dta \$150_V1_Match.dta	Data/Fortis Alberta/Phase 3 Survey Data	FALSE	FortisAlberta (2023)

Software Requirements

- Stata (code was last run on 02/27/2025 with Stata-MP version 18.0).
 - Packages are automatically installed in the do-file “Full Results Run.do”.
 - Packages include: unique, reghdfe (version 6.12.3 08aug2023), ftools (version 2.49.1 08aug2023), xlincom, coefplot, egenmore, carryforward, grc1leg, estout, grc1leg2, boottest
- R 4.4.2
 - Packages are installed in the file 0.libraries-install.R.
 - Packages include: plyr, tidyverse, haven, dplyr, lubridate, timeDate, ggplot2, ggExtra

Controlled Randomness

- Random seed is set at line 62 of program “0.Transformer limit Randomization.R”.
- Random seed is set at lines 540, 552, 556, 560, and 686 of program “0.Randomization.do”.
- Random seed is used in estimate bootstrap standard errors in lines 1393, 1401, 1411, 1419, 1430, 1438, 1464, 1480 of program “2. Regression Analysis – EV and Transformer Level.do”.

- Random seed is set at lines 326, 429, 544, 660, 776, 892, 1008, 1123, and 1239 of program “2.Transformer Upgrade Needs Calculation.do”.
- Random seed is set at line 214 of program “3.Attrition Reg Robustness.do”.
- Random seed is set at line 535, 1438, 1454, 2320, and 2336 of program “3.Cluster-Based Transformers Robustness.do”.

Memory and Runtime Requirements

Summary

Approximate time needed to reproduce the analyses on a standard 2025 desktop machine:

- 1-2 hours

Approximate storage space needed:

- 250 MB – 2 GB

Details

The code was last run on an Macbook Pro, 16-core CPU, M4 Max chip laptop with MacOS version Sequoia 15.3.1.

Description of programs/code

All the replication programs are available in “Replication File - Deidentified/Programs” folder. The code should be run in the following order:

1. STATA file that runs through all of the do-files: “Full Results Run.do”. Instructions of when to run R script “0.Transformer limit Randomization.R” is provided in the do-file.
2. R-File “4. Fig C.1 Hourly plot US data.R”.

Instructions to Replicators

- For each Stata program, edit “statapath” at the top of the code to set the directory to reflect the directory on your system.
- For each .R file, set the working directory in “directory” to reflect your system.
- Run all programs in the sequence described in “Full Results Run.do”.
- The results will be saved in the “Results” folder.

List of tables and programs

The provided code reproduces all tables and figures in the paper:

Figure/Table #	Program	Line Number	Output file (generates in Results folder)
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Figure 1	0. Distribution Transformer Illustrations.do	52	Figure1.pdf
Tables 1 and C2	1.Balance Pre-Treatment - Daily.do	338, 919	Table1.xlsx, TableC2.xlsx
Figure 2	1.Descriptive Statistics.do	471, 388	Figure2a.pdf, Figure2b.pdf
Tables 2, C3, C4, C5 (Column (1)) Figures 3, B1, B2, C2, C3	2. Regression Analysis - EV and Transformer Level.do	1070, 696, 1813, 1910, 416, 1170, 1227, 1262, 1297, 1330, 470, 505, 540, 573, 1648, 1708, 1775, 1675, 1735, 1803	Table 2.csv, Table C3.csv, Table C4.xlsx, Table C5.xlsx, Figure3a.pdf, Figure3b.pdf, FigureB1a.pdf, FigureB1b.pdf, FigureB1c.pdf, FigureB1d.pdf, FigureB2a.pdf, FigureB2b.pdf, FigureB2c.pdf, FigureB2d.pdf, FigureC2a.pdf, FigureC2b.pdf, FigureC2c.pdf, FigureC3a.pdf, FigureC3b.pdf, FigureC3c.pdf
Table C1	1.Extensive Margin Analysis.do	220	Table C1.xlsx
Table B1	1.Comparing Compliers and Non-Compliers - Daily.do	307	Table B1.xlsx

Table B2	3.Attrition Reg Robustness.do	777	Table B2.csv
Figure C1	4. Fig C.1 Hourly plot US data.R	117	FigureC1.pdf
Figures C4 and C5 Table C5 (Columns (2) and (3))	3.Cluster-Based Transformers Robustness.do	2209, 2169, 1343, 1303, 1476, 2355	FigureC4a.pdf, FigureC4b.pdf, FigureC5a.pdf, FigureC5b.pdf, Table C5.xlsx
Figure C6	2.Transformer Upgrade Needs Calculation.do	1356	FigureC6.pdf

References

FortisAlberta (2023). Electric Vehicle Smart Charging Pilot Data. Confidential Data. Request access by contacting FortisAlberta Privacy Officer (privacyofficer@fortisalberta.com).

Optiwatt (2024). United States Electric Vehicle Charging and Driving Data. Confidential Data. Request access by contacting Todd Touesnard (todd@optiwatt.com).